SHORT EXACT SEQUENCES OF GRADED PIECES ATTACHED TO SUBOBJECTS AND TO MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the first assertion, including injectivity, surjectivity and exactness in the middle; the closing sentence 'The two short exact sequence of (2) are special cases of the short exact sequence of (1).' is the author's own reduction, preserved verbatim, and it is justified because the objects involved carry exactly the induced or quotient filtrations introduced at the start.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

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Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Filtered objects, induced and quotient filtrations. Same chapter, the definitions of a filtered object, of a subobject with the induced filtration and of a quotient with the quotient filtration. Supplies the filtrations on X, A/X, on the kernel, coimage, image and cokernel of a morphism of filtered objects.
- The associated graded functor. Same chapter, the definition of the graded pieces gr^p and of the associated graded gr. Supplies the objects whose short exact sequences the lemma asserts.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category.*

(1) *Let A be a filtered object and $X \subset A$. Then for each p the sequence*

$$0 \to gr^p(X) \to gr^p(A) \to gr^p(A/X) \to 0$$

is exact (with induced filtration on X and quotient filtration on A/X).

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(2) *Let $f : A \to B$ be a morphism of filtered objects of $\mathcal{A}$. Then for each p the sequences*

$$0 \to gr^p(\mathrm{Ker}(f)) \to gr^p(A) \to gr^p(\mathrm{Coim}(f)) \to 0$$

and

$$0 \to gr^p(\mathrm{Im}(f)) \to gr^p(B) \to gr^p(\mathrm{Coker}(f)) \to 0$$

are exact.

Proof. We have $F^{p+1}X = X \cap F^{p+1}A$, hence map $\mathrm{gr}^p(X) \to \mathrm{gr}^p(A)$ is injective. Dually the map $\mathrm{gr}^p(A) \to \mathrm{gr}^p(A/X)$ is surjective. The kernel of $F^pA/F^{p+1}A \to A/X + F^{p+1}A$ is clearly $F^{p+1}A + X \cap F^pA/F^{p+1}A = F^pX/F^{p+1}X$ hence exactness in the middle. The two short exact sequence of (2) are special cases of the short exact sequence of (1). □